FLAT PULLBACK IS COMPATIBLE WITH COMPOSITION OF FLAT MORPHISMS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as three spans, lines 8-12 (the verbatim package together with the project's own reference, slogan and history display-environment definitions, including the author's commented-out older form of the reference environment), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). The lines 8-12 span is required because a counted statement or its surrounding text may use those display environments, which the project typesets as comments. Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Composition of morphisms and relative dimension. Morphisms, label morphisms-lemma-composition-relative-dimension-d, invoked at proof line 2300. Gives that the composite of the two flat morphisms is flat of relative dimension the sum, which is the first assertion of the lemma.
- Dimension of the inverse image of a cycle under a flat morphism. Same chapter chow.tex, label lemma-flat-inverse-image-dimension, invoked at proof line 2313. Shows that it suffices to compare coefficients for integral closed subschemes related by the two flat pullbacks, reducing the identity to a single coefficient comparison.
- Transitivity of the length formula for flat local ring maps. Algebra, label algebra-lemma-pullback-transitive, invoked at proof line 2327. Supplies the equality between the length of C over the expanded maximal ideal of A and the product of the length of C over the expanded maximal ideal of B with the length of B over the expanded maximal ideal of A, which is exactly the comparison of the two coefficients.

Source reference. Original lemma 02M2: sources/stacks-chow/tags.txt, source lines 3429–3429. Navigation pointer for \ref{algebra-lemma-pullback-transitive} used inside the counted span (line 2327); the argument that consumes it is retained in the span.

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Source reference. Original lemma 02R8: sources/stacks-chow/pinned-original-source.tex, source lines 2127–2127. Navigation pointer for \ref{lemma-flat-inverse-image-dimension} used inside the counted span (line 2313); the argument that consumes it is retained in the span.

Source reference. Original lemma 02NL: sources/stacks-chow/tags.txt, source lines 3484–3484. Navigation pointer for \ref{morphisms-lemma-composition-relative-dimension-d} used inside the counted span (line 2301); the argument that consumes it is retained in the span.

Source reference. Original reference 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Navigation pointer for \ref{situation-setup} used inside the counted span (line 2288); the argument that consumes it is retained in the span.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let X, Y, Z be locally of finite type over S. Let $f : X \to Y$ and $g : Y \to Z$ be flat morphisms of relative dimensions r and s. Then $g \circ f$ is flat of relative dimension $r + s$ and*

$$f^* \circ g^* = (g \circ f)^*$$

as maps $Z_k(Z) \to Z_{k+r+s}(X)$.

Proof. The composition is flat of relative dimension $r + s$ by Morphisms, Lemma 02NL. Suppose that

(1) $W \subset Z$ is a closed integral subscheme of δ-dimension k,
(2) $W' \subset Y$ is a closed integral subscheme of δ-dimension $k + s$ with $W' \subset g^{-1}(W)$, and
(3) $W'' \subset Y$ is a closed integral subscheme of δ-dimension $k + s + r$ with $W'' \subset f^{-1}(W')$.

We have to show that the coefficient n of $[W'']$ in $(g \circ f)^*[W]$ agrees with the coefficient m of $[W'']$ in $f^*(g^*[W])$. That it suffices to check the lemma in these cases follows from Lemma 02R8. Let $\xi'' \in W''$, $\xi' \in W'$ and $\xi \in W$ be the generic points. Consider the local rings $A = \mathcal{O}_{Z,\xi}$, $B = \mathcal{O}_{Y,\xi'}$ and $C = \mathcal{O}_{X,\xi''}$. Then we have local flat ring maps $A \to B$, $B \to C$ and moreover

$$n = \text{length}_C(C/\mathfrak{m}_A C), \quad \text{and} \quad m = \text{length}_C(C/\mathfrak{m}_B C)\text{length}_B(B/\mathfrak{m}_A B)$$

Hence the equality follows from Algebra, Lemma 02M2. □